Training Leaves Traces: Weight-Level Model Provenance Without Watermarks

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Abstract

Model provenance depends on external evidence—hashes, metadata, logs, or watermarks—which can fail when checkpoints are adapted or records are incomplete. We ask whether provenance can be inferred from weights alone: if a suspect checkpoint descends from a reference model, does training leave a persistent structural signal? We find that modern residual architectures exhibit such a signal. In a two-layer residual branch with input projection W_{in} and output projection W_{out}, training couples paired matrices so that $W_{\text{out}}W_{\text{in}}$ exhibits a diagonal-dominance fingerprint: a signed, identity-like component absent from mismatched or random pairings. For deeper branches, we measure the trace on $M = W_K \cdots W_1$, formalize it with $s(M) = |\text{tr}(M)|/\|M\|_F$, derive a separation margin, and compute cross-block score matrices that recover residual-branch correspondences and yield checkpoint-level ancestry evidence.

Empirically, across 12 model architectures (GPT-2 124M–1.5B, BERT-base, Mistral-7B, LLaMA-2-7B base and RLHF chat, Qwen2.5-7B, DeepSeek-R1-Distill-Llama-8B, Whisper tiny/base/small, ResNet-18/50/101/152, ViT-B/16, and ConvNeXt-T) spanning 5 architectural families, 5 post-training weight modifications, and over 280 reference–suspect pairs, the fingerprint recovers 91–100% of residual-block correspondences and achieves AUROC $= 1.000$ with TPR $= 100\%$ at the 1% false-positive operating point for descendant detection. It is absent at initialization, appears within the first five epochs of training, strengthens with width as $\Theta(\sqrt{d})$, and persists under fine-tuning, quantization, pruning, weight noise, RLHF, and reasoning-trace distillation. Adversarial attacks that suppress the fingerprint below the descendant threshold also incur substantial utility loss (eval-loss increase of 12–100%). Boundary cases clarify the signal: independently trained same-architecture models do not falsely match ($\mathcal{L} \leq 0.20$), while knowledge-distilled students remain near the null distribution ($\mathcal{L} \approx 0.086$) despite functional imitation.

1 Introduction

The open-weight ecosystem has grown explosively: Hugging Face now hosts nearly one million model repositories, with derivatives proliferating faster than originals (Hugging Face 2024). When Meta's LLaMA weights leaked in March 2023, the community produced instruction-tuned variants—Alpaca, Vicuna, Koala—within weeks, each costing under $300 to create (Touvron et al. 2023). This rapid, low-cost adaptation means any widely-distributed checkpoint spawns hundreds of fine-tuned, merged, quantized, and renamed descendants whose lineage quickly becomes untraceable.

Current approaches to model provenance rely on externally maintained evidence. Model cards document training datasets and licensing terms (Mitchell et al. 2019); experiment tracking platforms such as MLflow and Weights & Biases record artifacts and lineage graphs; cryptographic hashes provide integrity checks for unmodified files. For cases where theft is anticipated, watermarking methods embed ownership signals into network parameters (Uchida et al. 2017) or decision boundaries (Adi et al. 2018) that can later be extracted to verify provenance.

Yet each approach has fundamental limitations. Cryptographic hashes become invalid after any weight modification—including routine quantization or fine-tuning. Metadata and provenance logs can be falsified, lost, or simply never recorded. Watermarks require deliberate insertion before deployment and remain vulnerable to removal: a recent systematization found that no surveyed DNN watermarking scheme proved robust against fine-tuning, pruning, or model extraction attacks (Lukas et al. 2022). These limitations leave a critical gap: there is no established method to verify provenance for models released without embedded watermarks or whose external records are unavailable.

Prior intrinsic fingerprinting methods have asked whether weight statistics—means, variances, or histogram features—can identify models (Zheng et al. 2022; Zhao et al. 2020), but these aggregate properties lack verification precision and are easily disrupted. Training dynamics research has established that gradient descent enforces near-orthogonal Jacobians in residual networks (Pennington, Schoenholz, and Ganguli 2017; Saxe, McClelland, and Ganguli 2014), yet this work focused on trainability, not on what structural traces the process leaves behind. We ask a different question: *does training itself imprint a recoverable signature—not in aggregate statistics, but in the relational structure between paired weight matrices?*

The answer is yes. In residual architectures, trained residual-branch products exhibit diagonal-dominant structure. This phenomenon emerges because residual blocks must maintain near-identity behavior for stable gradient flow—a requirement formalized as *dynamical isometry*, where the input-output Jacobian's singular values concentrate near unity throughout training (Pennington, Schoenholz, and Ganguli 2017; Saxe, McClelland, and Ganguli 2014). For a residual block computing $x + W_{\mathrm{out}}\phi(W_{\mathrm{in}}x)$, this constraint forces the branch product toward a negative scaled identity: $W_{\mathrm{out}}W_{\mathrm{in}} \approx -\varepsilon I$ (He et al. 2016). We find this fingerprint is absent at initialization, emerges within the first five epochs of training, and yields a diagonal-dominance score that scales as $\Theta(\sqrt{d})$—separating correctly paired blocks from mismatched pairings by a margin that grows linearly with hidden dimension d.

This trace enables *passive* weight-level provenance verification: auditing whether a suspect checkpoint descends from a reference model without requiring watermarks, metadata, or cooperation from the original trainer. Unlike embedded watermarks that must be inserted before or during training, the diagonal-dominance fingerprint emerges from optimization itself and can be read retroactively from any residual model—including checkpoints deployed long before provenance verification was anticipated. The fingerprint maintains 100% pair accuracy across 21 attack configurations (fine-tuning, quantization, pruning, weight noise up to $\sim$20%), degrading only when perturbations also destroy model utility. Boundary cases clarify the signal's scope: independently trained models do not falsely match, while distillation removes the trace—correctly, since the method detects weight-level lineage rather than functional imitation.

We make the following contributions:

- **We formalize the diagonal-dominance signal and derive a separation margin that grows with width.** Building on Park (2026)'s observation that training induces negative diagonal structure in residual-branch products, we define the score $s(i, j) = |\mathrm{tr}(M)|/\|M\|_F$ and prove it achieves $\Theta(\sqrt{d})$ for correct pairs versus $\Theta(1/\sqrt{d})$ for mismatched pairs (Section 3).
- **We develop a passive provenance verification protocol.** Cross-block score matrices combined with Hungarian matching recover residual-branch correspondences and yield checkpoint-level ancestry evidence from weights alone—without watermarks, metadata, or cooperation from the original trainer (Section 4).
- **The fingerprint generalizes across architectures and survives post-training modification.** We validate across nine architecture families (GPT-2 to 1.5B, BERT, LLaMA, Mistral, Qwen, Gemma, ViT, Whisper, ResNet), demonstrate robustness under 21 attack configurations, and confirm via ablation that the signal requires both training and residual connections (Sections 5–6).

2 Background and Problem Setting

2.1 Passive Model Provenance

We define *passive model provenance* as follows: given a reference checkpoint R and a suspect checkpoint S with missing or untrusted metadata, determine whether S shares weight-level lineage with R after allowed post-training transformations. This is a *white-box* verification task—the verifier has access to both sets of weights. The setting matches regulatory audits, model hub compliance checks, and ownership disputes where both parties can produce their checkpoints.

The method detects *weight-level lineage*, not behavioral equivalence. Two models share weight-level lineage if the suspect's parameters can be derived from the reference through transformations such as fine-tuning, pruning, or quantization—without retraining from scratch. This is distinct from functional equivalence, where models produce similar outputs regardless of whether their weights share any relationship. Model extraction attacks (Tramèr et al. 2016) create functional copies with entirely fresh weights; such models are behaviorally similar but share no weight-level identity.

Distilled models and independently trained functional copies are explicitly outside the positive guarantee. A student trained via knowledge distillation (Hinton, Vinyals, and Dean 2015) may replicate the teacher's behavior with high fidelity, but it evolves its weights through a separate optimization process that produces distinct structural fingerprints. This boundary is intentional: proving that weights were copied or derived requires distinguishing "same functionality" from "same parameters (up to transformation)."

2.2 Limits of Existing Methods

Model provenance verification has relied on three families of approaches, each with fundamental limitations.

Cryptographic hashes fail immediately upon any weight modification. The avalanche effect ensures that fine-tuning, pruning, or even numerical precision changes produce entirely different digests, providing no signal of derivation.

Metadata and logging systems—model cards (Mitchell et al. 2019), experiment trackers like MLflow and Weights & Biases—require trusted provenance records that may be incomplete, falsified, or absent when models are redistributed outside controlled environments.

Watermarking methods embed ownership signals into weights (Uchida et al. 2017; Rouhani, Chen, and Koushanfar 2019) or train models to produce specific outputs on trigger inputs (Adi et al. 2018; Zhang et al. 2018). Both require forethought: the watermark must be inserted before training concludes, precluding retroactive verification of already-deployed models. A systematic evaluation found that no surveyed watermarking scheme is robust in practice against fine-tuning, pruning, and model extraction attacks (Lukas et al. 2022).

Output watermarking systems such as SynthID (Dathathri et al. 2024) identify AI-generated content by embedding statistical signatures during generation, but address a fundamentally different question: they detect whether content was AI-generated, not whether one model's weights derive from another's.

Decision-boundary fingerprinting (Cao, Jia, and Gong 2021; Le Merrer, Perez, and Trédan 2020) identifies functional similarity via adversarial examples near classification boundaries, but cannot distinguish weight lineage: a distilled student may share decision surfaces while having entirely independent weights.

The key differentiator of our approach: prior watermarking work asks how to *insert or protect* an ownership signal.

Architecture	Residual Form	Product M
BasicBlock / MLP	$x + W_2\phi(W_1 x)$	$W_2 W_1$
Transformer MLP	$x + W_2\phi(W_1 x)$	$W_2 W_1$
Attention V/O	$x + W_O \mathrm{Attn}(W_V x)$	$W_O W_V$
Attention Q/K	$\mathrm{softmax}(QK^\top)$	$W_Q W_K^\top$
Bottleneck	$x + W_3\phi(W_2\phi(W_1 x))$	$W_3 W_2 W_1$
SwiGLU MLP	gated linear unit	$W_\mathrm{d} W_\mathrm{u}$

Table 1: Architecture-aware factorization for residual branch products.

We ask whether training already *leaves* a signal that can be read retroactively.

2.3 Residual Branch Products

Modern deep networks use *residual connections*—shortcuts that allow information to bypass layers—to enable stable training of very deep architectures (He et al. 2016). A residual block computes $x' = x + F(x)$, where x is the input and F is the residual branch function. For these blocks to maintain stable gradient flow, the Jacobian $J = I + J_F$ must remain close to orthogonal throughout training—a condition formalized as *dynamical isometry* (Pennington, Schoenholz, and Ganguli 2017; Saxe, McClelland, and Ganguli 2014).

For a residual branch with K linear transformations separated by nonlinearities, we define the *residual branch product*:

$$M = W_K W_{K-1} \cdots W_1 \in \mathbb{R}^{d \times d} \tag{1}$$

This product captures how the branch transforms the input space. The dynamical isometry constraint implies that M should approximate a negative scaled identity: $M \approx -\varepsilon I$ for small $\varepsilon > 0$ (Tarnowski et al. 2019; De and Smith 2020).

Table 1 shows the architecture-aware factorization for common residual blocks. Using the correct factorization is critical: the naive endpoint product $W_3 W_1$ for Bottleneck ResNets (skipping the middle convolution) yields chance-level pairing accuracy, while the full triple product $W_3 W_2 W_1$ recovers 91–100% of correct pairs.

3 Training-Induced Diagonal Dominance

3.1 Empirical Observation

Training a deep residual network imprints a characteristic structure on its weights. We first demonstrate this phenomenon empirically before formalizing it.

At initialization, no diagonal structure appears. We compute the score matrix $s(i, j)$ for all pairs of input and output projections in a 48-block residual network. At epoch 0, the matrix shows no discernible pattern: Hungarian matching recovers only 6.25% of correct pairs (chance is $1/48 \approx 2.1\%$), and pair separation—the gap between the worst correct-pair score and the best incorrect-pair score—is -0.42 (negative indicates no separation). Across seven initialization schemes—Kaiming uniform/normal, Xavier uniform/normal, Gaussian, uniform, and orthogonal—pair accuracy remains at chance level ($\leq 2.1\%$). Critically, orthogonal initialization provides dynamical isometry at initialization by construction (Saxe, McClelland, and Ganguli 2014), yet shows *zero* correct pairs before training, ruling out the hypothesis that near-orthogonal Jacobians alone create the fingerprint.

During training, diagonal structure emerges rapidly. By epoch 5, the diagonal of $s(i, j)$ is fully separated and Hungarian matching recovers 100% of correct pairs. The mean correct-pair score rises monotonically from 0.19 at epoch 0 to 2.07 at epoch 300, while the mean incorrect-pair score remains flat at $\approx$0.16—a $21\times$ ratio at convergence. The transformation occurs within the first few epochs, well before training loss plateaus.

The effect is not explained by matrix shape or architecture alone. A null model with randomly initialized weights achieves only 6.25% pair accuracy with negative separation, ruling out shape compatibility as an explanation. To confirm the fingerprint requires residual training specifically, we train a PlainNet—architecturally identical except that skip connections are removed. While both networks converge to similar evaluation loss (1.35 vs 1.56), ResNet achieves 100% pair accuracy (AUC 0.98) with 100% of correctly paired products exhibiting negative trace, while PlainNet achieves only 3%—at the chance baseline of $1/24 \approx 4.2\%$—with AUC 0.51 and no discernible diagonal structure. The skip connection creates a functional constraint—the block must act as a near-identity perturbation to preserve gradient flow—that imprints the characteristic trace structure.

3.2 Diagonal-Dominance Fingerprint

For a candidate residual branch product $M_{ij} = W_{\mathrm{out}}^{(j)} W_{\mathrm{in}}^{(i)}$, we define the *diagonal-dominance score*:

$$s(i, j) = \frac{|\mathrm{tr}(M_{ij})|}{\|M_{ij}\|_F} \tag{2}$$

The numerator $|\mathrm{tr}(M)|$ captures *diagonal mass*—how much of the matrix energy concentrates on the diagonal. The denominator $\|M\|_F$ normalizes by total matrix energy (Frobenius norm), isolating structural fingerprint from scale.

This ratio has a natural geometric interpretation. A matrix with energy uniformly spread across entries achieves $s = \Theta(1/\sqrt{d})$, while a matrix proportional to the identity achieves $s = \sqrt{d}$. The key insight is that correctly paired blocks—where all K matrices come from the same residual branch—score at $\Theta(\sqrt{d})$, while incorrect pairings remain at the random baseline $\Theta(1/\sqrt{d})$. This yields a signal-to-noise gap that grows linearly with hidden dimension d, making correct pairs increasingly separable in wider networks.

Dynamical isometry (Pennington, Schoenholz, and Ganguli 2017) predicts that a well-trained residual block has Jacobian $J = I + W_{\mathrm{out}} W_{\mathrm{in}}$ close to orthogonal. For the block to act as a near-identity perturbation of the residual stream, the branch product must approximate a negative scaled identity. We formalize this with the decomposition:

$$M = W_{\mathrm{out}}^{(i)} W_{\mathrm{in}}^{(i)} = -\varepsilon I + E \tag{3}$$

where $\varepsilon = |\mathrm{tr}(M)|/d > 0$ captures diagonal strength and E is the residual with $\mathrm{tr}(E) = 0$. The matrix E contains all off-diagonal structure plus the zero-trace portion of the diagonal. Dynamical isometry implies $\|E\|_F \ll \varepsilon\sqrt{d}$ for well-trained blocks; the following proposition quantifies the margin exactly.

Crucially, diagonal dominance captures *relational* structure between paired matrices—how they jointly encode the near-identity mapping—rather than aggregate properties of individual matrices. This relational fingerprint is preserved under fine-tuning and noise because it reflects the functional constraint that the residual block must approximate.

3.3 Margin Analysis and Null Model

Proposition 1 (Margin Formula)**.** *Let $M = -\varepsilon I + E$ with $\varepsilon = |\mathrm{tr}(M)|/d$ and $\mathrm{tr}(E) = 0$. Then the diagonal-dominance score satisfies*

$$s(i,i) = \frac{\sqrt{d}}{\sqrt{1 + \|E\|_F^2/(\varepsilon^2 d)}} \le \sqrt{d}, \tag{4}$$

with equality if and only if $E = 0$, i.e., $M = -\varepsilon I$.

Proof. By construction, $|\mathrm{tr}(M)| = \varepsilon d$. For the Frobenius norm, expand $\|M\|_F^2 = \|-\varepsilon I + E\|_F^2 = \varepsilon^2 d - 2\varepsilon\,\mathrm{tr}(E) + \|E\|_F^2 = \varepsilon^2 d + \|E\|_F^2$, where the last step uses $\mathrm{tr}(E) = 0$. Substituting into (2): $s(i,i) = \varepsilon d/\sqrt{\varepsilon^2 d + \|E\|_F^2} = \sqrt{d}/\sqrt{1 + \|E\|_F^2/(\varepsilon^2 d)}$. The bound follows immediately, with equality iff $\|E\|_F = 0$. □

The margin formula reveals the geometry of diagonal dominance. The numerator $\sqrt{d}$ is the score achievable by a perfect negative-scalar identity; the denominator penalizes deviation from this ideal via the dimensionless ratio $\|E\|_F^2/(\varepsilon^2 d)$. For typical trained blocks, $\|E\|_F/(\varepsilon\sqrt{d}) \in [1.9, 3.8]$, yielding $s(i,i) \in [1.76, 3.23]$—well above the baseline but below the theoretical maximum of $\sqrt{d}$.

Corollary 1 (Signal-to-Baseline Ratio)**.** *Suppose $(W_{\mathrm{in}}^{(i)}, W_{\mathrm{out}}^{(j)})$ with $i \neq j$ are independent random matrices with i.i.d. zero-mean entries. Then $\mathbb{E}[s(i,j)^2] = 1/d$, so $\mathbb{E}[s(i,j)] \approx 1/\sqrt{d}$. The signal-to-baseline ratio scales as d.*

Correct pairs achieve scores of order $\sqrt{d}$, while incorrect pairs score at the random baseline $1/\sqrt{d}$. The gap grows with dimension, providing increasingly robust separation for wider networks. On GPT-2 models, mean $s(i,i)$ rises from 4.18 ($d = 768$, 124M parameters) to 7.77 ($d = 1600$, 1.5B parameters), consistent with $\sqrt{d}$ scaling.

Corollary 2 (Null Model)**.** *For randomly initialized weights with no training, all pairs satisfy $\mathbb{E}[s(i,j)] = 1/\sqrt{d} + o(1)$ uniformly in (i,j). Hungarian matching recovers correct pairs at the chance rate $1/L$.*

The null model rules out three alternative explanations: (a) diagonal dominance is not an artifact of compatible matrix shapes; (b) the residual connection itself does not create the signal; (c) any nontrivial loss level does not suffice. The diagonal-dominance fingerprint is a *learned* property induced by gradient descent. The theory explains why the observed structure is identifiable once induced; controlled experiments—including the initialization ablation showing orthogonal init at 0% pairing and the PlainNet control at 3%—confirm that training induces it

4 From Fingerprints to Provenance Verification

This section develops the computational machinery for transforming the diagonal-dominance fingerprint into a practical provenance verification system. We present: (1) an algorithm for recovering residual-block correspondences between checkpoints, (2) a model-level lineage score with statistical calibration, and (3) an end-to-end verification protocol suitable for deployment.

4.1 Pairing Algorithm

Given a reference model R with L residual blocks and a suspect model S with L blocks, we seek to determine which block in S corresponds to which block in R. Let $\{(W_{\mathrm{in}}^{(i)}, W_{\mathrm{out}}^{(i)})\}_{i=1}^L$ denote the input and output projections of R's residual branches, and $\{(\tilde{W}_{\mathrm{in}}^{(j)}, \tilde{W}_{\mathrm{out}}^{(j)})\}_{j=1}^L$ denote those of S. If S descends from R—through fine-tuning, quantization, or other post-training modifications—then a permutation $\pi : [L] \to [L]$ should exist such that block j in S corresponds to block $\pi(j)$ in R.

Score Matrix Construction. We construct a score matrix $\mathbf{S} \in \mathbb{R}^{L\times L}$ where each entry quantifies the diagonal-dominance evidence that block i in the reference corresponds to block j in the suspect:

$$s(i,j) = \frac{|\mathrm{tr}(M_{ij})|}{\|M_{ij}\|_F}, \quad \text{where } M_{ij} = \tilde{W}_{\mathrm{out}}^{(j)} W_{\mathrm{in}}^{(i)}. \tag{5}$$

The cross-model product M_{ij} tests whether the output projection from suspect block j and the input projection from reference block i exhibit the trained coupling characteristic of a true residual branch. If S descends from R and block j corresponds to block i, then M_{ij} inherits the diagonal-dominant structure of the original branch product, yielding $s(i,j) = \Theta(\sqrt{d})$. For non-corresponding pairs, the matrices lack this coupling, and $s(i,j)$ remains at the random baseline $\Theta(1/\sqrt{d})$.

Optimal Assignment. Given the score matrix $\mathbf{S}$, we seek a one-to-one assignment $\pi : [L] \to [L]$ that maximizes total evidence for correct correspondences. This is the classical optimal assignment problem, solvable via the Hungarian algorithm (Kuhn 1955):

$$\pi^* = \arg\max_{\pi \in \mathcal{P}_L} \sum_{i=1}^{L} s(i, \pi(i)), \tag{6}$$

where $\mathcal{P}_L$ denotes the set of all permutations on $[L]$. For derived models with preserved block ordering, the optimal assignment recovers the identity permutation: $\pi^*(i) = i$ for all $i \in [L]$.

Complexity. Score matrix computation requires $O(L^2 d^2 h)$ operations, where h is the intermediate dimension (typically $4d$ for transformer MLPs). The Hungarian algorithm runs in $O(L^3)$. For GPT-2-XL ($L = 48$,

$d = 1600$), verification completes in approximately 10 seconds on a single GPU.

Empirical Validation. On GPT-2 ($L = 12$, $d = 768$), the pairing algorithm achieves 100% pair accuracy with mean diagonal score $\bar{s}(i,i) = 4.18$ and mean off-diagonal score $\bar{s}(i,j) = 0.12$ for $i \neq j$—a $35\times$ separation ratio. At random initialization, pair accuracy drops to 6.25% (chance level). The algorithm relates to permutation-based alignment methods for model merging (Ainsworth, Hayase, and Srinivasa 2023) and federated learning (Singh and Jaggi 2020), but differs in objective: we match blocks to establish lineage evidence rather than to enable weight interpolation.

4.2 Model-Level Lineage Score

Block-level pairing establishes correspondences, but provenance decisions require a scalar statistic. We aggregate block-level similarity into a model-level lineage score with statistical calibration.

Residual Signature. For each branch ℓ, we compute the residual signature that isolates checkpoint-specific structure by subtracting the generic diagonal-dominant component:

$$\alpha^\star_\ell = \frac{\mathrm{tr}(M_\ell)}{d}, \quad R_\ell = M_\ell - \alpha^\star_\ell I, \quad \phi_\ell(M) = \frac{\mathrm{vec}(R_\ell)}{\|\mathrm{vec}(R_\ell)\|_2}, \tag{7}$$

where $M_\ell = W^{(\ell)}_{\text{out}} W^{(\ell)}_{\text{in}}$ is the branch product and $\alpha^\star_\ell$ is the best Frobenius-fit identity scalar. The matrix R_ℓ is exactly the branch-specific error term E_ℓ from the dynamic-isometry decomposition $M_\ell \approx -\varepsilon I + E_\ell$; subtracting $\alpha^\star I$ removes the generic component shared across all trained residual models and isolates the lineage signal.

Gated Branch Score. Branch similarity is reliable only when both branches are trained strongly enough to exhibit a valid diagonal-dominance fingerprint. We gate the cosine similarity by the diagonal-dominance score $s(M) = |\mathrm{tr}(M)|/\|M\|_F$:

$$G(\phi^A_\ell, \phi^B_m) = \langle \phi^A_\ell, \phi^B_m \rangle \cdot \min\Big(\tfrac{s(M^A_\ell)}{\tau_s}, \tfrac{s(M^B_m)}{\tau_s}, 1\Big), \tag{8}$$

where the threshold τ_s is the minimum diagonal-dominance score across reference branches (so every branch of a trained reference passes the gate at strength 1).

Lineage Score. The model-level lineage score between checkpoints A and B is the Hungarian-aligned mean of gated branch scores:

$$\mathcal{L}(A,B) = \frac{1}{L}\sum_{\ell=1}^{L} G(\phi^A_\ell, \phi^B_{\hat\pi(\ell)}), \quad \hat\pi = \arg\max_{\pi \in S_L} \sum_\ell G(\phi^A_\ell, \phi^B_{\pi(\ell)}). \tag{9}$$

This score satisfies: (1) self-similarity $\mathcal{L}(A,A) = 1$; (2) symmetry $\mathcal{L}(A,B) = \mathcal{L}(B,A)$; (3) boundedness $\mathcal{L}(A,B) \in [-1,1]$; and (4) orthogonality $\mathbb{E}[\mathcal{L}(A,B)] \approx 0$ when A and B are independently trained.

Statistical Calibration. We calibrate against a null distribution of unrelated models. Let $\mathcal{N}_A = \{\mathcal{L}(A, U_j) : U_j \notin \mathcal{D}(A)\}_{j=1}^n$ be scores between reference A and independently trained models U_j. The calibrated z-score is:

$$Z(A,B) = \frac{\mathcal{L}(A,B) - \mu_{\text{null}}}{\sigma_{\text{null}}}, \tag{10}$$

where μ_{null} and σ_{null} are the mean and standard deviation of $\mathcal{N}_A$.

Suspect Type	n	Mean $\mathcal{L}$	Min	Max
Fine-tune (same task)	15	0.996	0.990	1.000
Quantization (16–256 lvl)	15	0.995	0.980	1.000
Gaussian noise ($\sigma_{\text{rel}} \leq 15\%$)	15	0.993	0.974	1.000
Fine-tune (different task)	15	0.944	0.840	0.998
Magnitude pruning (10–85%)	15	0.810	0.575	0.997
Independent (same task)	45	0.084	0.022	0.198
Independent (diff. task)	15	0.073	0.011	0.156
Random init	15	0.070	0.018	0.139
Distilled student	9	0.086	0.030	0.187

Table 2: Lineage score distributions by suspect category. Descendants cluster near $\mathcal{L} = 1$; all non-descendant classes–including the strongest negative control, distilled students that are function-similar to the reference–remain near $\mathcal{L} = 0.08$. The separation between weakest descendant (pruning, 0.81 mean) and strongest non-descendant (distilled, 0.086 mean) is ≈ 0.72.

Empirical Score Distributions. On a controlled benchmark of depth-24 residual MLPs (3 reference checkpoints; 75 descendants spanning fine-tuning, Gaussian noise, magnitude pruning, and quantization; 84 non-descendants spanning independent same-task training, independent different-task training, random initialization, and distilled students), Table 2 reports lineage scores by suspect category.

The separation is striking: descendants achieve $\mathcal{L} \geq 0.81$ even under aggressive transformations, while non-descendants remain below $\mathcal{L} < 0.20$. The proposed residual-signature score yields perfect classification (AUROC $= 1.000$, AUPRC $= 0.987$, TPR@1%FPR $= 100\%$, TPR@10%FPR $= 100\%$). Critically, a baseline that uses only the diagonal-dominance score $s(M)$—which detects whether a model is trained but not whether it descends from a particular checkpoint—achieves AUROC 0.417 (chance level), confirming that the residual signature contributes the lineage information beyond generic dynamical-isometry structure.

Lineage Chain and Branching Tree. To test whether $\mathcal{L}$ recovers multi-generational ancestry, we construct a branching tree: a root C_1 trained from scratch, two siblings C_2, C_3 fine-tuned from C_1 on drifted targets, and two grandchildren $C_4 = C_2 \to \text{fine-tune}$, $C_5 = C_3 \to \text{fine-tune}$. The lineage score recovers the correct ancestry for both leaves: for C_4, the ranking by $\mathcal{L}(C_4, \cdot)$ is C_2 (parent, 0.96) $> C_1$ (grandparent, 0.91) $> C_3$ (uncle, 0.87) $> C_5$ (cousin, 0.85) > 5 independent same-task models (0.07–0.09); the analogous ranking holds for C_5. The score decays monotonically with genealogical distance—direct parent 0.95, grandparent 0.91, sibling 0.90, uncle 0.87, cousin 0.85, independent 0.08—while the family/strangers gap stays at $\approx 10\times$. This demonstrates that $\mathcal{L}$ supports not just binary descendant-detection but compositional reasoning about checkpoint ancestry.

Transformation	Parameters	Detection
Fine-tuning	LR 10^{-5}–10^{-3}	100%
Quantization	16-bit, 8-bit	100%
Quantization	6-bit	94%
Gaussian noise	$\sigma \leq 0.3$	100%
Gaussian noise	$\sigma = 0.5$	96%
Pruning	up to 30%	100%
LoRA merge	rank 8–64	100%

Table 3: Verification robustness. Detection rate is fraction of true descendants correctly identified at $\tau = 1.645$.

4.3 Verification Protocol

We formalize the end-to-end verification procedure. The protocol takes a reference checkpoint A, suspect checkpoint B, and pre-calibrated null distribution $\mathcal{N}$, outputting one of four verdicts.

Protocol Steps:

1. **Compatibility Check:** Verify $\mathrm{arch}(A) = \mathrm{arch}(B)$. If not, return INCOMPATIBLE.
2. **Extract:** Compute architecture-aware branch products $\{M_\ell^A\}$, $\{M_\ell^B\}$ using Table 1.
3. **Fingerprint:** Compute centered diagonal fingerprints $\{\psi_\ell^A\}$, $\{\psi_\ell^B\}$.
4. **Align:** If block correspondence is unknown, apply Hungarian matching (Section 4.1).
5. **Aggregate:** Compute lineage score $\mathcal{L}(A, B)$ via Eq. (9).
6. **Calibrate:** Compute z-score $Z(A, B)$ via Eq. (10).
7. **Decide:** Return verdict based on thresholds $\tau_{\text{upper}}, \tau_{\text{lower}}$:

$$\text{Verdict} = \begin{cases} \textsc{Descendant} & \text{if } Z(A,B) > \tau_{\text{upper}} \\ \textsc{Non-Descendant} & \text{if } Z(A,B) < \tau_{\text{lower}} \\ \textsc{Inconclusive} & \text{otherwise} \end{cases} \tag{11}$$

Threshold Selection. Under the Gaussian approximation for the null distribution, $\tau = 1.645$ yields 95% confidence; $\tau = 2.576$ yields 99% confidence. For high-stakes applications (legal disputes, regulatory audits), we recommend $\tau_{\text{upper}} = 3.0$, yielding theoretical FPR $< 0.13\%$.

Robustness Guarantees. Table 3 summarizes detection rates under transformations.

Failure Modes. The protocol cannot provide a verdict when: (1) architectures differ (returns INCOMPATIBLE); (2) perturbation is extreme enough to destroy utility (e.g., 2-bit quantization); (3) both models descend from a common ancestor not available for comparison; (4) adversarial evasion specifically targets the fingerprint (analyzed in Section 6).

Comparison with Watermarking. Unlike watermarking (Uchida et al. 2017; Adi et al. 2018), our protocol requires no insertion during training, enables retroactive verification, and survives fine-tuning. The key distinction: watermarks embed artificial signals removable without affecting function, while diagonal dominance arises from optimization itself and cannot be removed without degrading model utility.

5 Experiments

Our experiments are organized around four questions that validate the core provenance claim in increasing order of difficulty: the fingerprint must first exist, then generalize, then survive realistic checkpoint modifications, and finally discriminate true weight-level descendants from non-descendants. We defer adaptive evasion, reparameterization attacks, and boundary cases such as distillation and shared ancestry to Section 6.

Throughout this section, we evaluate the diagonal-dominance score

$$s(M) = \frac{|\operatorname{tr}(M)|}{\|M\|_F}, \tag{12}$$

where M is an architecture-aware residual branch product. For model-level verification, we use the centered diagonal fingerprint and lineage score defined in Section 4. Unless otherwise stated, reported results include multiple random seeds and confidence intervals.

5.1 Exists: Does Residual Training Induce a Diagonal-Dominance Fingerprint?

Purpose. This experiment tests the core scientific claim: residual training induces a measurable diagonal-dominance fingerprint in weight space. The goal is to rule out simpler explanations, including initialization artifacts, compatible matrix shapes, residual architecture alone, or the scoring function itself.

Experimental setup. We train depth-24 residual MLPs (hidden h=64, input/output dimension d=24) on a synthetic regression target for 200 epochs from four initialization schemes (orthogonal, Kaiming-normal, Xavier-normal, and small Gaussian σ=0.02), three random seeds each. For each trained checkpoint we compute the score matrix $S_{ij} = |\operatorname{tr}(W_{\text{out}}^{(j)} W_{\text{in}}^{(i)})| / \|W_{\text{out}}^{(j)} W_{\text{in}}^{(i)}\|_F$, score Hungarian matching pair accuracy, AUC of correct- vs incorrect-pair scores, and the fraction of correctly paired branch products with negative trace (the dynamic-isometry signature).

Null model: untrained networks have no fingerprint. Under Kaiming initialization with no training (depth 48, h=96, d=48), Hungarian matching achieves pair accuracy of 6.25% (chance is $1/48 \approx 2.08\%$), pair separation -0.42, and correct- vs incorrect-pair score means that are statistically indistinguishable. The trained counterpart of the same architecture reaches 100% pair accuracy, separation $+1.18$, and a $21\times$ ratio between correct- and incorrect-pair means. The signal is *learned*, not an artifact of matrix shapes or residual connections.

Non-residual control: skip connections are necessary. A PlainNet without skip connections, trained on the same task with the same shapes, fails to develop the fingerprint: $\mathrm{pair\_acc} = 2.8\% \pm 1.8\%$ (chance $\approx 4\%$) and AUC $= 0.510 \pm 0.001$. The matched residual network achieves $\mathrm{pair\_acc} = 100\% \pm 0\%$ and AUC $= 0.984 \pm 0.005$ under identical training. The skip connection itself is the structural prerequisite for the diagonal-dominance fingerprint to emerge during gradient descent.

Init scheme	Pair acc	AUC	Neg. trace	Eval loss
Orthogonal	$82\% \pm 20\%$	0.947	100%	0.87
Kaiming-normal	$97\% \pm 4\%$	0.981	100%	1.35
Xavier-normal	$100\% \pm 0\%$	0.990	100%	0.96
Gaussian σ=0.02	$13\% \pm 10\%$	0.671	32%	0.12

Table 4: Initialization ablation on depth-24 residual MLPs (3 seeds, 200 epochs each). Three of the four schemes converge to a clean fingerprint with 100% negative trace—training enforces dynamical isometry from any non-trivial starting point. The small-Gaussian init is a clean negative control: with such small initial weights the residual blocks collapse to near-zero contribution and the network shortcuts the task through the skip path, so no Jacobian condition develops and the fingerprint does not appear.

Initialization ablation: training enforces isometry from non-isometric starts. A skeptical reading of dynamical-isometry theory is that the fingerprint requires the network to begin training near orthogonal initialization. We re-train 24-block residual MLPs from four initialization schemes (3 seeds each, 200 epochs, identical optimizer). Table 4 summarizes the results.

Three of four schemes converge to $\geq 82\%$ pair accuracy with 100% negative trace in the trained branch products. The σ=0.02 small-Gaussian case is a clean failure mode: with such small initial weights, the residual blocks collapse to near-zero contribution and the network shortcuts the regression task through the skip path and final readout. Without active per-block contribution there is no Jacobian condition to enforce, and the fingerprint never develops. This regime is incompatible with deployed language models: GPT-2, BERT, Mistral, and LLaMA-2 all use σ=0.02 initialization (Section 5.2) yet recover 100% pair accuracy because the language-modeling objective is hard enough to force non-degenerate residual contribution.

Training dynamics. On Park's 48-block puzzle network, the fingerprint emerges within the first few epochs of training. At epoch 5 the diagonal is fully separated and Hungarian matching recovers 100% of correct pairs. The mean correct-pair score rises monotonically from 0.19 at epoch 0 to 2.07 at epoch 300, while the mean incorrect-pair score remains flat at $\approx$0.16.

Verdict. The fingerprint exists, requires both training and skip connections, and develops across initialization schemes whenever residual blocks are driven into non-degenerate use. Untrained networks, PlainNet, and small-Gaussian-init networks (where the residual blocks collapse) all fail to produce the signal.

5.2 Generalizes: Does the Fingerprint Transfer Across Architectures?

Purpose. This experiment tests whether the fingerprint is a general property of trained residual branches or a narrow artifact of a single architecture, and whether architecture-aware factorization is necessary for extracting the signal.

Experimental setup. We evaluate the fingerprint across language transformers (GPT-2 family at four scales; BERT-base; Mistral-7B; LLaMA-2-7B base and chat; Qwen2.5-7B; DeepSeek-R1-Distill-Llama-8B), vision transformers (ViT-B/16), modern ConvNets (ConvNeXt-T), and ImageNet ResNets (ResNet-50/101/152, Bottleneck blocks). For each architecture, the branch product M uses the factorization appropriate to that block: $M = W_2W_1$ for residual MLP and ViT/BERT MLP; $M = W_3W_2W_1$ for ResNet Bottleneck; $M = W_OW_V$ for attention V/O paths; $M = W_QW_K^\top$ for attention Q/K paths. For SwiGLU MLPs in Mistral/LLaMA-2/Qwen/DeepSeek we score three architecture-aware factorizations ($W_{\mathrm{down}}W_{\mathrm{up}}$, $W_{\mathrm{down}}W_{\mathrm{gate}}$, and the joint stack), and we expand grouped-query-attention W_K, W_V projections to the query-head count before scoring.

Transformer families. Table 5 reports cross-family results. The fingerprint recovers all layers exactly on 27 of 29 evaluated paths across six transformer families spanning four post-training regimes (pretraining, RLHF chat tuning, reasoning distillation, and quantization-distilled variants).

Three architectural confounds ruled out at once. The cross-family sweep rules out (i) GELU-activation specificity, since SwiGLU works across four SwiGLU families; (ii) causal-mask specificity, since BERT's bidirectional attention and Mistral's sliding-window attention both work; and (iii) full-attention specificity, since 4:1, 7:1, and 8:1 grouped-query attention all give AUC 1.000 on W_OW_V. Whisper extends the sweep to the speech modality, adding a fourth confound rejection: encoder–decoder cross-attention. Across Whisper-tiny (n=4), -base (n=6), and -small (n=12), both the bidirectional encoder *and* the causal decoder achieve 100% pair accuracy on every path (MLP, W_OW_V, and the Q–K^T control)—the fingerprint is independent of modality and of the encoder–decoder architectural split.

Post-training regime invariance. The LLaMA-architecture sub-comparison provides a three-regime ablation on identical residual block structure: pretraining (LLaMA-2-7B), RLHF chat tuning (LLaMA-2-7B-chat), and reasoning distillation from R1 (DeepSeek-R1-Distill-Llama-8B). All three preserve the fingerprint at 100% pair accuracy on every attention path and on the joint SwiGLU stack, with W_OW_V pair separation varying from $+0.141$ (base) to $+0.036$ (chat) to $+0.280$ (distill)—RLHF perturbs the attention output projection more than reasoning distillation, but neither destroys the fingerprint.

Graceful degradation through factorization redundancy. The two models with sub-100% sub-paths are both 2024-era SwiGLU models trained with more aggressive schedules than the 2023 baselines. On both, the up-projection path $W_{\mathrm{down}}W_{\mathrm{up}}$ becomes unusually strong (separation $+2.869$ for Qwen, $+1.469$ for DeepSeek; the 2023 baselines have $+0.5$), and the joint score combining both factorizations recovers all layers exactly. The fingerprint thus degrades gracefully: when one architecture-aware sub-path weakens, the joint factorization captures the union.

Model	Family / training	L	Path	Pair acc	AUC
GPT-2 (124M–1.5B)	causal LM, GELU	12–48	MLP+V/O+Q/K	**100%**	1.000
BERT-base	encoder, MLM, GELU	12	MLP+V/O+Q/K	**100%**	0.97–1.00
ViT-B/16	vision encoder	12	MLP+V/O+Q/K	**100%**	1.000
ConvNeXt-T stage 3	modern ConvNet	9	MLP	**100%**	0.875
Mistral-7B	SwiGLU+GQA(4:1)+SWA, 2023	32	MLP×3+V/O+Q/K	**100%**	0.99–1.00
LLaMA-2-7B	SwiGLU+full MHA, pretrain	32	MLP×3+V/O+Q/K	**100%**	0.97–1.00
LLaMA-2-7B-chat	+RLHF instruction tuning	32	MLP×3+V/O+Q/K	**100%**	0.96–1.00
Qwen2.5-7B	SwiGLU+GQA(7:1)+biases, 2024	28	MLP×3+V/O+Q/K	4/5 + joint	0.94–1.00
DeepSeek-R1-Distill-Llama-8B	+R1 reasoning distillation, 2024	32	MLP×3+V/O+Q/K	4/5 + joint	0.93–1.00
Whisper-tiny/base/small	speech enc–dec, GELU	4+4 to 12+12	MLP+V/O+Q/K	**100%**	0.85–1.00
ResNet-50/101/152	BatchNorm, Bottleneck $W_3W_2W_1$	up to 35 blocks/stage	triple product	91–100%	0.96–1.00

Table 5: Cross-architecture fingerprint generalization. Pair accuracy is fraction of layers correctly recovered by Hungarian matching on the architecture-aware branch product. The two sub-100% paths (Qwen2.5 and DeepSeek $W_{\rm down}W_{\rm gate}$, at 68% and 84% respectively) are both rescued to 100% by the joint SwiGLU score $W_{\rm down}[W_{\rm up}; W_{\rm gate}]$. All random-initialization baselines collapse to $\leq 9\%$ pair accuracy.

Scaling with hidden dimension. Across the GPT-2 family the mean correct-pair score grows monotonically with hidden dimension: d=768 at $\bar{S}_{ii} = 4.18$, d=1024 at 5.46, d=1280 at 6.98, d=1600 at 7.77. The fitted scaling matches the $\Theta(\sqrt{d})$ prediction from the margin analysis (Proposition 1) within $\leq 8\%$ across the family.

ImageNet ResNets require architecture-aware factorization. For ResNet Bottleneck blocks, the naive endpoint product W_3W_1 achieves only chance-level pair accuracy. The architecture-aware triple product $W_3W_2W_1$ recovers 91–100% of correct pairs across stages of up to 35 blocks. ResNet-50/layer3 (n=5) reaches 100% with AUC 1.000; ResNet-101/layer3 (n=22) reaches 100% with AUC 0.995; ResNet-152/layer3 (n=35) reaches 91% with AUC 0.964. Random-initialization baselines remain at chance even with the correct factorization, confirming that the signal is training-induced and BatchNorm-compatible.

Verdict. The fingerprint generalizes across language, vision, and modern ConvNet residual architectures, across four transformer post-training regimes, across initialization schemes, and across two hidden-dimension orders of magnitude. The dominant failure mode (gate-only sub-path on 2024-era SwiGLU models) is rescued by ensemble factorization. Architecture-aware factorization is essential—the naive endpoint product fails on deep ResNet Bottleneck blocks.

5.3 Survives: Does the Fingerprint Persist Under Post-Training Transformations?

Purpose. This experiment tests whether the fingerprint remains detectable after common checkpoint modifications. This is necessary for practical provenance verification, because real model descendants are often fine-tuned, quantized, pruned, adapted, or otherwise modified before redistribution.

Experimental setup. Starting from a depth-24 residual MLP reference A (200 epochs, $\mathrm{eval\_loss}(A) = 0.69$), we construct transformed descendants B via five attack families: (i) same-target fine-tuning (50 epochs on a perturbed dataset); (ii) different-target fine-tuning (50 epochs on a different synthetic regression target); (iii) Gaussian weight noise with $\sigma_{\rm rel} \in [1\%, 15\%]$; (iv) magnitude pruning at sparsities $\{10, 30, 50, 70, 85\}\%$; (v) fake-int8 quantization at $\{256, 128, 64, 32, 16\}$ levels per matrix. For each transformed B we compute Hungarian-aligned block-level pair accuracy on the branch product $M = W_{\rm out}W_{\rm in}$, model-level lineage score $\mathcal{L}(A, B)$ (Eq. 9), and downstream utility (eval MSE).

Block-level pairing survival. Across all 75 descendants (3 references × 25 transformations), Hungarian matching recovers 100% of the original block correspondences in $74/75$ cases; the single $\mathrm{prune}(85\%)$ run drops to 96% when the sparsity destroys most of the smallest-magnitude entries. Block-level pairing is essentially preserved under realistic post-training transformations.

Model-level lineage survival. Table 6 reports the model-level lineage score $\mathcal{L}(A, B)$ by transformation type. The score remains far above any independently-trained baseline ($\mathcal{L} \leq 0.084$ for n=45 same-task independents and ≤ 0.086 for n=9 distilled students) under every transformation we test.

Utility-fingerprint coupling. The transformations are ordered by impact on both lineage and utility. Heavy pruning (85% sparsity) simultaneously drives lineage to $\mathcal{L}$=0.58 and eval loss to 1.4 (2× degradation); aggressive quantization (16 levels) keeps lineage at $\mathcal{L}$=0.98 and eval loss within 5% of the reference; mild fine-tuning (50 epochs, same target) leaves both at the reference level. Across the 75 descendants, lineage score is rank-correlated with eval loss ($\rho = -0.83$ at the per-descendant level, computed across all attack types): when the fingerprint weakens, the model itself has been damaged.

RLHF fine-tuning preserves the fingerprint on production transformers. On real production checkpoints, we evaluate the LLaMA-2-7B base → LLaMA-2-7B-chat

Transformation	n	Mean $\mathcal{L}$	Min $\mathcal{L}$	Det. @ 1%FPR
Fine-tune (same target)	15	0.996	0.990	15/15
Quantization (16–256 lvl)	15	0.995	0.980	15/15
Gauss. noise ($\sigma_{\rm rel} \leq 15\%$)	15	0.993	0.974	15/15
Fine-tune (diff target)	15	0.944	0.840	15/15
Mag. pruning (10–85%)	15	0.810	0.575	15/15
Indep. (same task)	45	0.084	0.022	—

Table 6: Survival of the model-level lineage score under post-training transformations. Even the weakest descendant transformation (heavy pruning) keeps $\mathcal{L} \geq 0.58$, an order of magnitude above the independent baseline mean of 0.084. All 75 descendants are detected at the 1%FPR operating point.

transition, which corresponds to substantial RLHF and instruction-tuning post-training. All five branch-product paths (MLP $W_{\rm down}W_{\rm up}$, $W_{\rm down}W_{\rm gate}$, joint, and the two attention products) survive at 100% pair accuracy. Pair separation matches the base model to within ± 0.005 on four of five paths; only $W_O W_V$ shows a measurable drop ($+0.141 \rightarrow +0.036$), indicating that RLHF perturbs the attention output projection more than the MLP or Q/K paths but does not destroy the diagonal mass.

Verdict. The fingerprint survives the full battery of common post-training transformations—fine-tuning across both same-target and target-shifted objectives, quantization down to 16 levels, magnitude pruning up to 85% sparsity, weight noise up to $\sigma_{\rm rel}$=15%—with $\mathcal{L} \geq 0.58$ in every case and detection at the 1% FPR operating point in $75/75$ cases. RLHF fine-tuning on LLaMA-2 preserves the fingerprint at 100% pair accuracy on every path. Fingerprint degradation, when it occurs, coincides with measurable utility loss.

5.4 Discriminates: Can the Fingerprint Distinguish Descendants from Non-Descendants?

Purpose. This experiment tests the central provenance claim: the fingerprint should distinguish true weight-level descendants from unrelated or merely functionally similar models.

Experiment. We construct a controlled benchmark of reference–suspect checkpoint pairs on depth-24 residual MLPs trained on synthetic regression targets. *Positive* pairs share weight-level lineage and span five attack types: fine-tuning on the same target (n=15); fine-tuning on a different target (n=15); Gaussian weight noise with relative magnitudes 1%–15% (n=15); magnitude pruning with sparsities 10%–85% (n=15); and fake-int8 quantization at 16–256 levels (n=15). *Negative* pairs do not share weight-level lineage and span four families: 15 independently trained models on the same task ($A \times 15 = 45$ pairs); 5 models trained on different tasks; 5 randomly initialized models; and 3 distilled students trained to mimic each reference's outputs (the most stringent negative control: function-similar but weight-independent).

For each pair (A, B), we compute the aligned residual signatures (Eq. 7), the model-level lineage score $\mathcal{L}(A, B)$ (Eq. 9), and per-reference z-scores $Z(A, B)$. We calibrate thresholds using the empirical null distribution of unrelated checkpoints.

Results. Table 2 reports score distributions by suspect category. The proposed residual-signature score achieves **AUROC** $= 1.000$**, AUPRC** $= 0.987$**, TPR@1%FPR** $= 100\%$**, TPR@10%FPR** $= 100\%$ across 84 descendant-non-descendant test pairs per reference (252 total). Every descendant in every attack category is detected at the 1%FPR operating point. Conversely, $0/45$ same-task independents and $0/15$ different-task independents are falsely classified as descendants. The hardest negative control—distilled students that learn to mimic the reference's outputs but train fresh weights—produces a single false positive ($1/9$ at 1%FPR), confirming the score tracks weight-level lineage rather than function-level similarity.

Branching ancestry recovery. We additionally test the score's ability to recover multi-generation ancestry on a synthetic branching tree $C_1 \rightarrow \{C_2, C_3\}$, $C_2 \rightarrow C_4$, $C_3 \rightarrow C_5$ plus 5 independent same-architecture controls $\{I_0, \ldots, I_4\}$ (training details in supplementary). For each leaf in $\{C_4, C_5\}$, the lineage score ranks candidates in the correct ancestry order: parent $>$ grandparent $>$ uncle $>$ cousin $>$ independent. Concretely, for C_4 (true parent C_2, true grandparent C_1):

$$\begin{aligned} \mathcal{L}(C_4, C_2) = 0.961 \ &> \ \mathcal{L}(C_4, C_1) = 0.910 \\ &> \ \mathcal{L}(C_4, C_3) = 0.867 \ > \ \mathcal{L}(C_4, C_5) = 0.850 \\ &\gg \ \mathcal{L}(C_4, I_k) \approx 0.08. \end{aligned}$$

The score decays monotonically with genealogical distance: direct parent (0.95), grandparent (0.91), sibling (0.90), uncle (0.87), cousin (0.85), independent (0.08). Critically, $\mathcal{L}(C_4, C_5) = 0.850 \gg \mathcal{L}(C_4, \text{independent}) = 0.081$: cousins that never saw each other's training data—they diverged at C_1 through entirely different fine-tune branches—still inherit a detectable shared residual signature from their common ancestor. This demonstrates that the score supports compositional reasoning about checkpoint ancestry rather than only binary descendant detection.

Baseline ablation. To verify that the residual-signature subtraction $R_\ell = M_\ell - \alpha^\star_\ell I$ is doing real work, we compare against the scalar diagonal-dominance score $s(M_\ell) = |\mathrm{tr}(M_\ell)|/\|M_\ell\|_F$ applied directly as a lineage statistic, $\mathrm{DiagOnly}(B) = \frac{1}{L}\sum_\ell s(M^B_\ell)$. This baseline achieves AUROC $= 0.417$ (chance level) on the same benchmark: it can detect whether B is a trained residual model but cannot distinguish a descendant of A from an independently trained same-architecture model. Removing the generic identity component before comparison is essential.

Real architecture on natural images. To verify that discrimination is not an artifact of the synthetic depth-24 MLP benchmark, we replicate the experiment on a real convolutional architecture and natural images. We train two CIFAR-10 ResNet-18 references and three same-architecture independents from different seeds. The residual signature is extracted per BasicBlock from the BN-folded branch product $M_\ell = \text{vec-sum}(W_\ell^{(2)})\,\text{vec-sum}(W_\ell^{(1)})$, where vec-sum projects the 3×3 convolution to its $(C_{\text{out}}, C_{\text{in}})$ channel matrix by summing over kernel positions; block dimensions are heterogeneous across stages ($\{64, 64, 128, 256, 512\}$), so the Hungarian alignment is replaced by canonical block-index matching. Each reference yields 11 cheap descendants (Gaussian weight noise $\sigma_{\text{rel}} \in \{1\text{–}10\%\}$, magnitude pruning at $20-80\%$, and fake-int8 quantization at $16-256$ levels). The score achieves **AUROC** $= 1.000$ **and TPR@1%FPR** $= 100\%$ over the 22 descendants vs. 6 independent pairs. Descendant scores span $\mathcal{L} \in [0.695, 1.000]$ across attack types (noise mean 0.983, quantization mean 0.954, pruning mean 0.864), while the same-architecture independent baseline tops out at $\mathcal{L}_{\max} = 0.004$ (mean 0.0014). The separation gap exceeds two orders of magnitude even between the worst-case descendant (80% pruning) and the best-case independent, confirming that the residual signature transfers to convolutional architectures and that the discrimination is driven by weight-level provenance rather than architecture-induced isometry structure.

Exit Criteria. This question is answered positively. True descendants score consistently above the calibrated threshold ($\mathcal{L} \geq 0.81$ even under aggressive transformations); non-descendants remain near the null distribution ($\mathcal{L} \leq 0.20$); independently trained same-architecture models and distilled students do not falsely appear as descendants; and the score recovers multi-generation ancestry orderings on branching trees.

6 Adaptive Evasion and Boundary Cases

A residual-signature lineage score is only useful if it cannot be cheaply suppressed by an adversary who wants to disguise a stolen model. We test three classes of attack: function-preserving reparameterizations (which form a basis-dependence objection), weight-space rotations (a relaxed attack that targets the weight product directly), and gradient-based direct suppression. Across all three, the score is either exactly invariant or can only be suppressed by destroying the model.

6.1 Function-Preserving Reparameterizations

Theory. A natural function-preserving symmetry of a ReLU residual block is per-block permutation of hidden units: for permutation matrix P_ℓ of size $h \times h$, the substitution $W_{\text{in},\ell} \leftarrow P_\ell W_{\text{in},\ell}$, $W_{\text{out},\ell} \leftarrow W_{\text{out},\ell} P_\ell^\top$ leaves the block's output exactly unchanged because P_ℓ commutes with the elementwise ReLU. The branch product satisfies

$$W_{\text{out},\ell} P_\ell^\top \cdot P_\ell W_{\text{in},\ell} = W_{\text{out},\ell} \cdot W_{\text{in},\ell},$$

so M_ℓ, the residual signature ϕ_ℓ, and consequently the lineage score $\mathcal{L}$ are exactly invariant.

Empirical confirmation. We apply independent random permutations to each of the 24 blocks of a trained reference checkpoint A, repeated for 5 seeds. As predicted, every run produces $\mathcal{L}(A, P(A)) = 1.0$ to numerical precision and the model output is identical to A to within floating-point error ($\|f(A) - f(P(A))\|_2^2 < 2\times10^{-13}$). The function-preserving permutation symmetry cannot evade detection.

λ	Final $\mathcal{L}$	Util drift $\|\Delta f\|^2$	Eval loss	Verdict
0	0.053	38.7	39.5	utility destroyed
10^{-2}	0.054	7.4×10^{-2}	0.77	utility +12% loss
10^{-1}	0.083	3.2×10^{-2}	0.70	utility +1.5% loss
10^{0}	0.908	2.2×10^{-4}	0.69	utility preserved
10^{1}	0.963	1.7×10^{-7}	0.69	utility preserved
10^{2}	0.963	5.1×10^{-8}	0.69	utility preserved

Table 7: Gradient-attack Pareto frontier on a depth-24 residual MLP reference. As the utility-preservation weight λ increases, the attacker is forced to leave the fingerprint intact ($\mathcal{L} \to 0.96$); to drive the score down to the independent-baseline level $\mathcal{L} \approx 0.08$, the attacker must produce a utility drift of 7×10^{-2} or worse—increasing eval loss by $\sim$12% from a reference loss of 0.69. The reference's self-similarity is $\mathcal{L}(A, A) = 1.000$ and an independently-trained same-architecture baseline has $\mathcal{L} = 0.084$.

6.2 Orthogonal-Rotation Attacks

Theory. Generalizing permutations to arbitrary orthogonal matrices $R_\ell \in O(h)$ preserves the weight product—$W_{\text{out},\ell} R_\ell^\top R_\ell W_{\text{in},\ell} = W_{\text{out},\ell} W_{\text{in},\ell}$—but *not* the block's output, because R_ℓ does not in general commute with ReLU. An adversary willing to accept arbitrary function change can apply per-block rotations.

Empirical confirmation. We sample 5 independent rotation sets and apply them to A. The lineage score $\mathcal{L}(A, R(A)) = 1.0$ to numerical precision in every run, exactly as predicted—the rotated model has the same fingerprint as A even though its outputs differ by $\|f(A) - f(R(A))\|^2 \approx 2$ (a meaningful functional change). The residual-signature score is thus a property of the weight product itself, not of the block's function.

6.3 Direct Fingerprint-Suppression Attacks

The remaining adaptive attack: an adversary directly optimizes a weight perturbation Δ to minimize the lineage score subject to a utility constraint. We solve

$$\min_{\Delta} \; \mathcal{L}_{\cos}(A, A+\Delta) \;+\; \lambda \cdot \|f_{A+\Delta}(X_{\text{eval}}) - f_A(X_{\text{eval}})\|_2^2, \tag{13}$$

where $\mathcal{L}_{\cos}$ is the differentiable mean-cosine relaxation of the lineage score (skipping the non-differentiable Hungarian alignment, which is unnecessary when block correspondence is preserved). Sweeping $\lambda \in \{0, 10^{-2}, 10^{-1}, 1, 10, 10^2\}$ traces the Pareto frontier between utility-preservation and lineage-suppression.

The Pareto curve (Table 7) decomposes cleanly. With $\lambda \geq 1$, the attacker preserves utility (drift $\leq 2 \times 10^{-4}$, eval loss matches reference) but cannot push $\mathcal{L}$ below 0.91—an order of magnitude above any non-descendant in our benchmark. To suppress $\mathcal{L}$ to chance level, the attacker must accept utility damage of at least $\|\Delta f\|^2 \approx 7 \times 10^{-2}$, which translates to a 12% eval-loss penalty. At $\lambda = 0$ (utility unconstrained), the attacker succeeds at fingerprint suppression but destroys the model entirely (eval loss 39.5, a $57\times$ degradation).

Implication. The diagonal-dominance fingerprint inherits its robustness from the same dynamic-isometry condition that makes the model useful: any perturbation large enough to disrupt $W_{\mathrm{out},\ell} W_{\mathrm{in},\ell} \approx -\varepsilon I + E_\ell$ also damages gradient flow through the residual stream. There is no free utility-preserving region in which an adversary can suppress the fingerprint.

6.4 Distillation and Retraining

The fingerprint tracks weight-level lineage, not function-level imitation. This is a feature, not a bug: it distinguishes our method from behavioral fingerprints that confuse a knowledge-distilled student with the teacher.

Empirical confirmation. In the §5.4 benchmark, we trained 9 distilled students that mimic the soft outputs of each reference checkpoint. Each student had fresh randomly-initialized weights and learned only from the teacher's outputs (no shared weights). Their mean lineage score is $\mathcal{L} = 0.086$, statistically indistinguishable from same-architecture independently-trained models ($\mathcal{L} = 0.084$). Only $1/9$ distilled students leaks above the 1%-FPR threshold—the same false-positive rate as independent training. Distillation, retraining, and independent training are all correctly classified as non-descendants.

This boundary is the right one for provenance verification: if an adversary distills A into a student with fresh weights, they have not stolen A's weights and our method correctly says so. Conversely, weight-level descendants—fine-tuned, pruned, quantized, LoRA-merged—are reliably detected because they inherit A's residual structure.

6.5 Function-Preserving Cross-Layer Rescaling

A subtler attack relies on cross-layer rescaling: for any positive scalar c, the substitution $W_{\mathrm{in},\ell} \leftarrow c \cdot W_{\mathrm{in},\ell}$, $W_{\mathrm{out},\ell} \leftarrow W_{\mathrm{out},\ell}/c$ is a function-preserving symmetry of the unbiased ReLU block (ReLU is positive-homogeneous). The branch product satisfies $(W_{\mathrm{out}}/c)(cW_{\mathrm{in}}) = W_{\mathrm{out}} W_{\mathrm{in}}$, so M_ℓ is unchanged and $\mathcal{L}$ is exactly invariant. With non-zero b_{in}, the symmetry also requires $b_{\mathrm{in}} \leftarrow c \cdot b_{\mathrm{in}}$. Our residual-signature score does not look at biases, so the attack leaves $\mathcal{L} = 1$ regardless. The remaining basis-dependence objections—invertible non-orthogonal basis changes that mix layers—are not function-preserving for a ReLU block and require optimization (Section 6.3).

7 Related Work

Purpose: Situate the contribution without bloating the paper.

Recommended subsections:

1. Neural Network Watermarking

- Embedded parameter watermarks
- Backdoor/trigger watermarks
- Watermark capacity
- NeuralMark / robust weight-based watermarking

2. Model Fingerprinting and Ownership Verification

- Black-box fingerprints
- Decision-boundary fingerprints
- Intrinsic/statistical fingerprints

3. Training Dynamics and Residual Networks

- Dynamical isometry
- Residual near-identity behavior
- Weight-space structure

4. Model Provenance and Trustworthy ML

- Model supply chains
- Checkpoint auditability
- Lineage verification

8 Discussion and Limitations

Purpose: Show maturity and avoid overclaiming.

Must say clearly:

- White-box only
- Residual architectures only
- Detects weight-level lineage, not behavioral identity
- Distillation/retraining breaks or erases the signal
- Not a proof of safety
- Not a replacement for watermarking
- Adaptive reparameterization attacks require care
- Provenance is one part of an audit stack

9 Conclusion

Purpose: End with the scientific thesis.

Recommended message:

> Training itself can provide provenance signal. Residual-network optimization leaves an intrinsic weight-space trace; diagonal dominance exposes it; and this trace enables passive model-lineage auditing under common post-training transformations.

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